Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} *: Drain Source Voltage.*

V_{GS} *: Gate Source Voltage.*

V_{TH} *: Threshold Voltage.*

U_t *: Thermal Voltage.*

n *: Weak Inversion Slope factor.*

g_m*: Transconductance.*

Solutions

1.

a) State two advantages and two disadvantages of encoding analogue signals in time rather than amplitude.

Advantages:

- Supply independence
- Spike domain coding can be event driven resulting in low power (triggers processing) [2 points]

Disadvantages

- Specialised circuits are required for encoding.
- Data is quantised in time, and therefore susceptible to phase noise. [2 points]

[4]

b) The circuit shown in Figure 1.1 is based on two phases, a reset phase when RST = 0 and a readout phase where RST = VDD. The input voltage is

i. State which domain the output V_{out} is encoded in and what limits the dynamic range of the encoded output signal.

[2]

ii. Assuming that M0 is biased in weak inversion, describe the behavior of V_a in both phases. Derive the rate of change of V_a during the readout phase. Considering that the switching point of the inverters is at $\frac{V_{DD}}{2}$, derive the transfer function between the encoded output and V_{in}. Define all constants used in MOSFET transconductance equations.

[6]

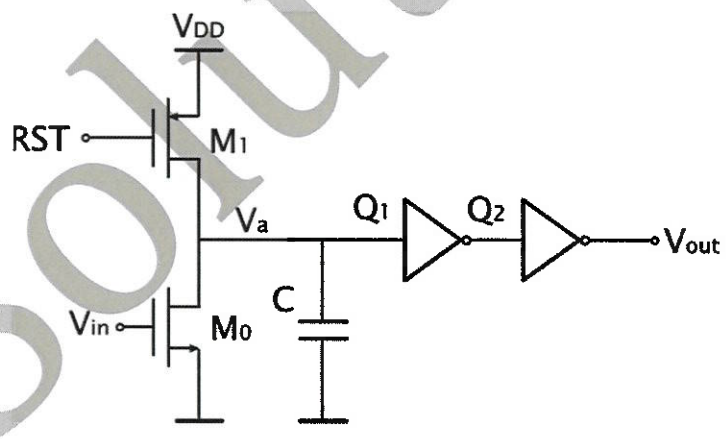


Figure 1.1

[Application of taught theory]

i. The circuit encodes the input in the time domain. Dynamic range is limited by how finely we can resolve the time signals, which affects the LSB. Jitter normally limits this.

ii.

In the reset phase, M1 is on and V_a is charged up to V_{DD}.

In the readout phase, m1 is off so the C discharges through M0. M0 is biased in weak inversion so the current flowing the device is

Solutions

$$I_{DS,0} = I_0 \exp\left(\frac{V_{in}}{nU_t}\right)$$

where $U_t = \frac{kT}{q} \approx 26\,mV$ at ambient temperature and n is the subthreshold slope.

The discharge current for the capacitor C is

$$I_{DS,0} = C\frac{dV_a}{dt}$$

Hence V_a discharges with a rate $I_{DS,0}/C$.

$$\frac{dV_a}{dt} = \frac{I_0}{C}\exp\left(\frac{V_{in}}{nU_t}\right)$$

The inverters quantify the discharge as a pulse, encoding the input voltage in time. Assuming that the switching point of the inverters is at $\frac{V_{DD}}{2}$ (and that the discharge is linear), the pulse width is given by

$$\tau = \frac{V_{DD}}{2}\frac{C}{I_0}\exp\left(-\frac{V_{in}}{nU_t}\right)$$

c) For the cascade of amplifiers shown in Figure 1.1, derive the output referred noise and explain what factors can be optimized to improve the signal to noise ratio. [8]

Mos Noise: $v_n^2 = \frac{K_w}{I_D^p}\cdot\Delta f + \frac{K_f}{A}\cdot\ln\left(\frac{f_h}{f_l}\right)$

Distributed gain: $G_i = \prod_{k=i}^{k=M} g_k$

Total output Noise: $v_{no}^2 = \sum_{i=1}^{i=M} v_{ni}^2 G_i^2$

Final Equation: $v_{no}^2 = \sum_{i=1}^{i=M} v_{ni}^2 G_i^2 = \sum_{i=1}^{i=M}\left(n_i\frac{K_w}{(I_i/n_i)^p}\cdot\Delta f + n_i\frac{K_f}{(A_i/n_i)}\cdot\ln\left(\frac{f_h}{f_l}\right)\right)G_i^2$ [6 points]

- Increasing Power and Area reduce the noise
- If we are willing to compute slowly (Δf and fh/fl are both small) then we can keep output noise low without consuming a lot of power. [2 points]

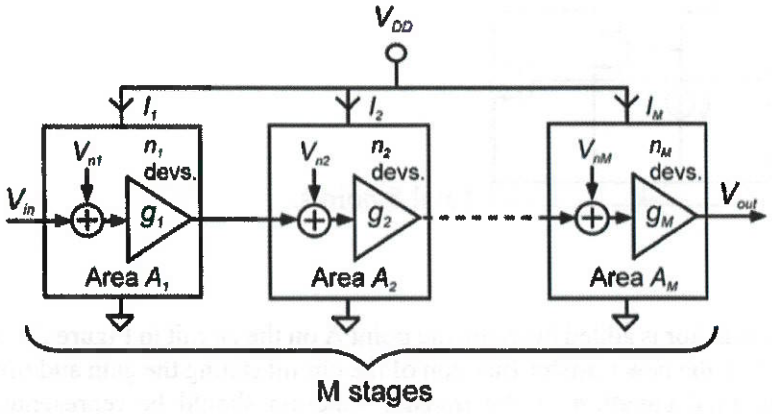


Figure 1.1: Cascade of amplifiers

Solutions

2. This question concerns translinear Circuits.

a) Give two limitations when using MOSFETs in translinear circuits.

Drawbacks:
Currents need to be limited to below 100nA. [1]
Temperatures must be equal to get rid of thermal voltage terms. [1]

[2]

b) Figure 2.1 shows a translinear circuit.

i. Derive the function of the circuit using the translinear principle. I_z is the output and I_x is the input.

Squaring Circuit from notes:

$I_1.I_2 = I_3.I_4$ and therefore $I_Z = \frac{(I_X)^2}{I_0}$ Total 2 points.

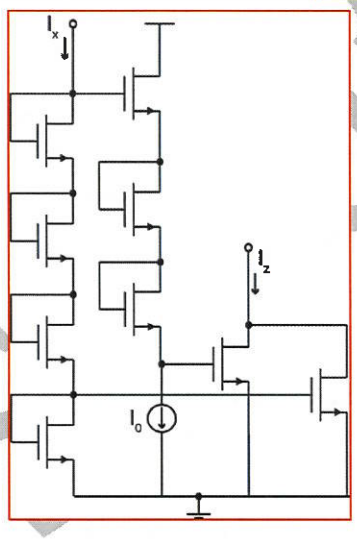
[2]

ii. Show how, with the addition of a few transistors, the circuit can be modified to calculate the function:

$$I_Z = \frac{(I_X)^4}{I_0^3} - I_x$$

New circuit derivation:

$I_Z = \frac{(I_X)^4}{I_0^3} - I_x$



Total 5 points.

[4]

iii. A capacitor is added between the point A on the circuit in Figure 2.1 and ground. Derive the new transfer function of the circuit stating the gain and time constant. The final equation of the transfer function should be represented in the s-domain.

Derivation from theory:
Capacitor current:

Solutions

$$I_c = CnU_t \cdot \frac{\dot{I}_Z}{I_Z}$$

$$I_Z = \frac{(I_X)^2}{I_0 + I_c}$$

Combining:

$$I_z = \frac{I_x^2}{I_o + CnU_t \cdot \frac{\dot{I}_z}{I_z}}$$

$$I_z I_o + CnU_t \cdot \dot{I}_z = I_x^2$$

Taking laplace transform:

$$I_z I_o + CnU_t \cdot sI_z = I_x^2$$

$$I_z(I_o + CnU_t \cdot s) = I_x^2$$

$$I_z = \frac{I_x^2}{I_o + CnU_t \cdot s} = \frac{I_x^2 / I_0}{1 + CnU_t \cdot s / I_0}$$

Gain = I_x^2 / I_0

Time constant: $\tau = CnU_t / I_0$

[4]

c) Figure 2.2 shows a translinear circuit capable of computing the trigonometric function sin(πy) through the approximation $\frac{y-y^3}{1+y^2}$.

i. Express the currents I_3 and I_4 in terms of I and I_X.

[4]

New Derivation

$$\left[\frac{I - I_X}{2}\right]^2 II_4 = I_3 \left[\frac{I + I_X}{2}\right]^2$$

$$\frac{I_4}{I_3} = \left[\frac{I+I_X}{I-I_X}\right]^2 \text{ and } I_3+I_4=I$$

$$I_4 = \frac{I(I + I_X)^2}{(I - I_X)^2 + (I + I_X)^2}$$

$$I_3 = \frac{I(I-I_X)^2}{(I-I_X)^2+(I+I_X)^2}$$

[4]

ii. Express the differntial output current I_2-I_1 and show that when I_X = yI then I_2-I_1 = $I\frac{y-y^3}{1+y^2}$

[4]

$$I_2 = \frac{I - I_X}{2} + \frac{I(I + I_X)^2}{(I - I_X)^2 + (I + I_X)^2}$$

$$I_1 = \frac{I + I_X}{2} + \frac{I(I - I_X)^2}{(I - I_X)^2 + (I + I_X)^2}$$

$$I_2 - I_1 = \frac{I^2 I_X - I_X^3}{I^2 + I_X^2}$$

$$I_2 - I_1 = I\frac{y - y^3}{1 + y^2}$$

Solutions

[4]
[Total 8 points]

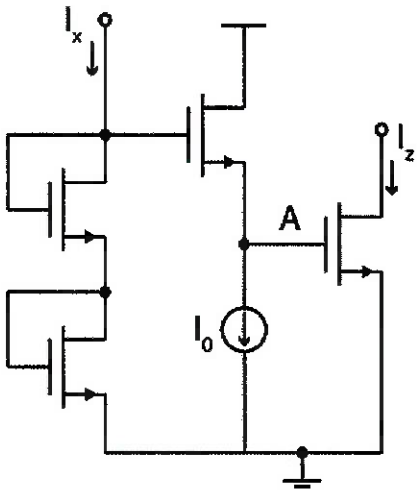


Figure 2.1

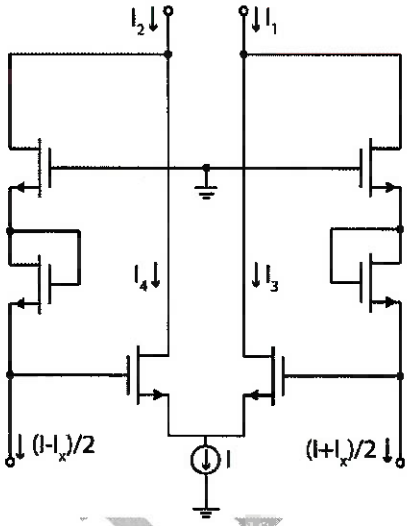


Figure 2.2

Solution

3.

a) Explain the principle of companding, illustrating how the dynamic range changes through the signal processing chain.

[Bookwork]
Definition [2]
Diagram [2]

[4]

b) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function is defined as $I_{out} = g(V_c) = I_0 \sinh(V_c/nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f(x) composed of just I_{in}, I_1 and I_{cosh}, whereby $I_{cosh} = I_0 \cosh(V_c/nU_t)$.

New derivation:

$$Y = I_{out} = I_0 \sinh(V_c / nU_t) = g(V_C)$$

$$\frac{d(g(V_c))}{dV_C} = \frac{dY}{dV_C} = \frac{dI_{out}}{dV_C} = \frac{I_0}{nU_t}\cosh(V_C / nU_t) = \frac{I_{Cosh}}{nU_t}$$

$$f(x) = I_C = X\tau C\left(\frac{dY}{dV_C}\right)^{-1} = X\tau C\frac{nU_t}{I_{\cosh}} = X\frac{I_1}{I_{\cosh}} \qquad X = I_{in}, \quad I_1 = \tau \cdot CnU_t$$

$$I_C = f(I_{in}) = \frac{I_{in}I_1}{I_{\cosh}}$$

[6]

c) The transfer function of a biquad filter is defined in state space representation by the following equations:

$$\dot{X}_1 = \frac{-\omega_0}{Q}X_1 - \omega_0 X_2 + \omega_0 U_1$$

$$\dot{X}_2 = \omega_0 X_1 - \omega_0 U_2$$

$$Y = X_1$$

whereby Y is the output and U_1 and U_2 are the inputs and X_1 and X_2 are the states.

i. By using the mappings below show how these linear equations can be mapped to non-linear log-domain design equations which represent a low pass filter. State any assumptions you make.

Solutions

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U_2 = I_{U2} \exp\left(\frac{V_{U2}}{nU_t}\right)$$

New derivation from Theory:
For low pass filter U1=0.

$$\frac{I_1}{nU_t}\exp\left(\frac{V_1}{nU_t}\right)\dot{V}_1 = -\frac{\omega_0}{Q} I_1 \exp\left(\frac{V_1}{nU_t}\right) + \omega_0 I_2 \exp\left(\frac{V_2}{nU_t}\right)$$

$$\frac{I_2}{nU_t}\exp\left(\frac{V_2}{nU_t}\right)\dot{V}_2 = -\omega_0 I_1 \exp\left(\frac{V_1}{nU_t}\right) + \omega_0 I_U \exp\left(\frac{V_U}{nU_t}\right)$$

$$C_1\dot{V}_1 = -\frac{C_1\omega_0 nU_t}{Q} + C_1\omega_0 nU_t \frac{I_2}{I_1}\exp\left(\frac{V_2 - V_1}{nU_t}\right)$$

$$C_2\dot{V}_2 = -C_2\omega_0 nU_t \frac{I_1}{I_2}\exp\left(\frac{V_1 - V_2}{nU_t}\right) + C_2\omega_0 nU_t \frac{I_U}{I_2}\exp\left(\frac{V_U - V_2}{nU_t}\right)$$

Let C1=C2=C and define $I_\omega = C\omega_0 nU_t$

$$C\dot{V}_1 = -\frac{I_\omega}{Q} + I_\omega \frac{I_2}{I_1}\exp\left(\frac{V_2 - V_1}{nU_t}\right)$$

$$C\dot{V}_2 = -I_\omega \frac{I_1}{I_2}\exp\left(\frac{V_1 - V_2}{nU_t}\right) + I_\omega \frac{I_U}{I_2}\exp\left(\frac{V_U - V_2}{nU_t}\right)$$

A suitable mapping:

$$I_1 = I_2 = I_\omega$$

$$I_U = I_0$$

Final equations:

$$C\dot{V}_1 = -\frac{I_\omega}{Q} + I_\omega \exp\left(\frac{V_2 - V_1}{nU_t}\right)$$

$$C\dot{V}_2 = -I_\omega \exp\left(\frac{V_1 - V_2}{nU_t}\right) + I_0 \exp\left(\frac{V_U - V_2}{nU_t}\right)$$

$$Y_1 = I_\omega \exp\left(\frac{V_1}{nU_t}\right)$$

$$Y_2 = I_\omega \exp\left(\frac{V_2}{nU_t}\right)$$

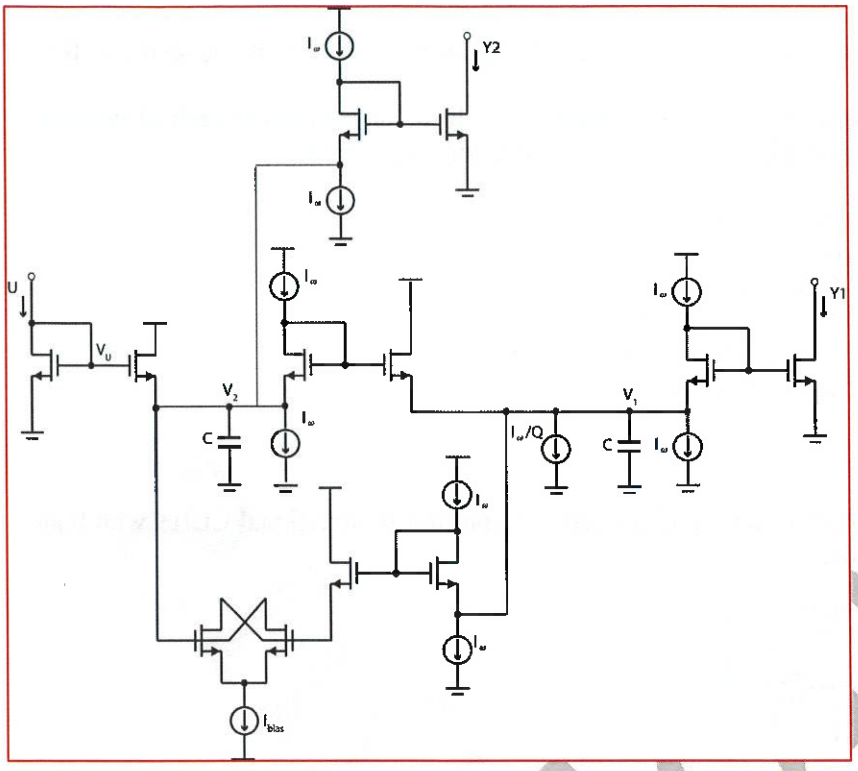
[5]

ii. With these log-domain design equations, sketch a schematic of the final log domain filter using weak inversion MOS transistors.

[5]

A Solution using the above equations:

Solutions



Solutions

4.

a) Figure 4.1 shows a conventional second-generation positive current conveyor, CCII+.

i. Explain its operation principle and describe how impedances at each of its three ports, X, Y, Z, differ from a standard operational amplifier.

[bookwork]

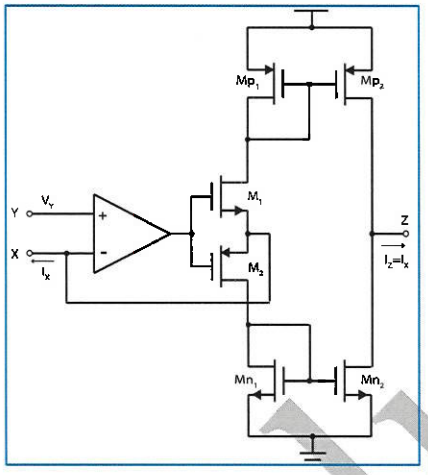
- Current-voltage characteristics
  $V_X=V_Y$, $I_Z=\pm I_X$, $I_Y=0$
  $Z_Y \rightarrow \infty$, $Z_X \rightarrow 0$, $Z_Z \rightarrow \infty$ [2]
- Voltage-follower between Y-input and X-output
- Current-follower between X-input and Z-output [1]

Plus a comparison with Op-amp terminals [1]

[4]

ii. Draw the circuit of a CMOS implementation of a bi-directional CCII+ with high output impedance.

Schematic:



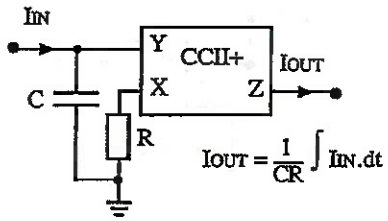
[3]

We can use cascaded current mirrors to increase output impedance at the expense of dynamic range. [1]

[4]

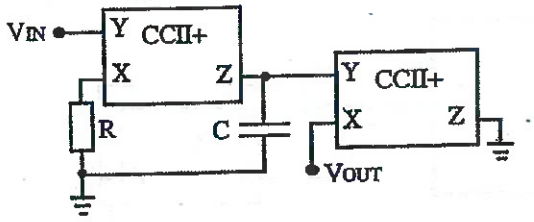
iii. Using the current conveyor in Figure 4.1, draw the schematics of a current mode integrator and a voltage mode integrator, each with a time constant, τ =1s. You may assume R = 100KΩ.

Current integrator [2]:



Voltage integrator [2]:

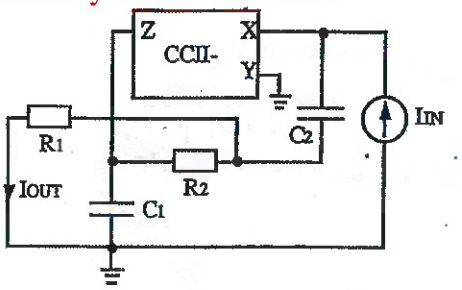
Solutions



[4]

b) Figure 4.2 shows a voltage mode Sallen key filter. Using the adjoint principle transform this to it's current mode equivalent using CCII+ blocks.

Sallen Key Circuit with CCII-:



[2]

Student also needs to realize the CCII- block required can be implemented using two CCII+ blocks. [2]

[4]

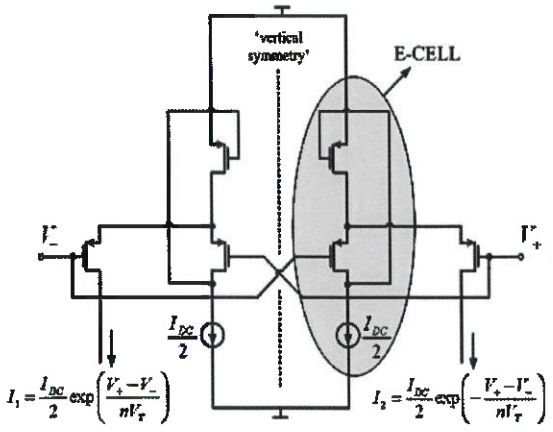
c) Sketch a current mode circuit that can implement the following hyperbolic functions:

$$I_1 - I_2 = I_{DC}\sinh\left(\frac{V_1 - V_2}{nU_t}\right)$$

$$I_1 + I_2 = I_{DC}\cosh\left(\frac{V_1 - V_2}{nU_t}\right)$$

[4]

[bookwork]



$$I_1 - I_2 = I_{DC}\sinh\left(\frac{V_1 - V_2}{nU_t}\right)$$

$$I_1 + I_2 = I_{DC}\cosh\left(\frac{V_1 - V_2}{nU_t}\right)$$

Solutions

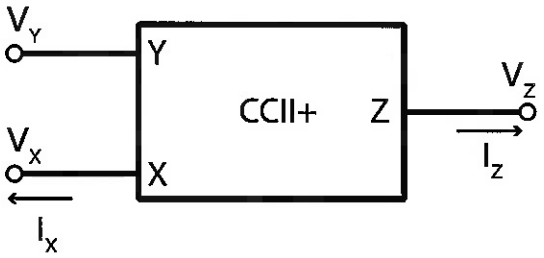


Figure 4.1

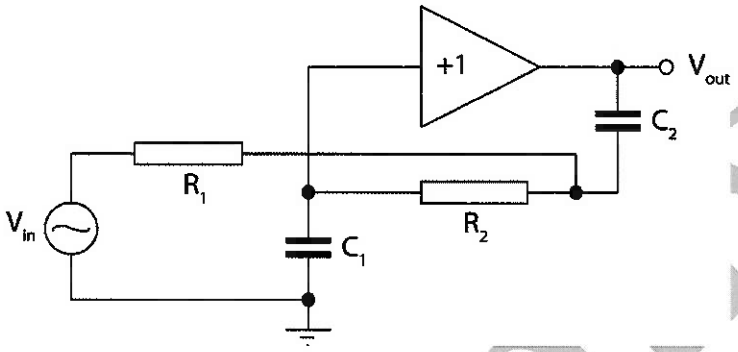


Figure 4.2

Solutions

5. This question relates to discreet time integrators.

a) Figure 5.1 shows a discreet time RC integrator.

i) Derive the time constant of integration. Assuming $C_1 = C_2$, select a suitable clock frequency to give a time constant $\tau = 1\mu s$.

Switches M1, M2 and C1 for a resistor R, and time constant $\tau = RC_2$ where

$$R = \frac{1}{C_1 f_{CK}}$$

$\tau = C_2 / C_1 * f_{ck}$ which results in f_{ck}=1MHz

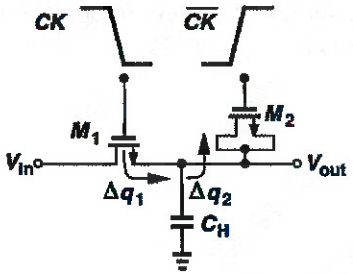
[2]

ii) State four possible non-idealities in switches. Describe a way to cancel the one which is usually dominant and demonstrate mathematically.

There are four non-idealities in switches :

- Charge injection.
- Clock feedthrough.
- Sampled noise.
- Leakage current.

Charge injection is dominant in most cases. Its effect can be cancelled by adding a dummy transistor with source and drain connected :



For cancellation, we need the charge injected from M1 to match the charge required by M2:

$$\Delta q_1 = \Delta q_2$$

Assuming that half the charge is deposited on C_H,

$$\Delta q_1 = \frac{W_1 L_1 C_{ox}}{2}(V_{CK} - V_{in} - V_{th1})$$

and that the charge required for M2 is

$$\Delta q_2 = W_2 L_2 C_{ox}(V_{CK} - V_{in} - V_{th2})$$

Charge injection is cancelled if $W_1 L_1 = 2W_2 L_2$ (or any combination involving width and length separately).

[4]

iii) For the circuit Figure 5.1, when CK goes low all of the channel charge, Q_1, of transistor M_1 is injected into C_1. This causes an error voltage ΔV_{C1} in C_1 defined by:

$$\Delta V_{C1} = \frac{Q_1}{C_1}$$

Then transistor M_2 turns ON. The channel charge of M_2 is split equally between C_1 and C_2. Estimate the total amount of charge injection and therefore error voltage ΔV_{C2} from switches

Solutions

M_1 and M_2 which appears on the output capacitor C_2 after CK goes high again, that is after one integration cycle. [4]

[New application of taught theory]

Charge in Q_1 is injected into C_1 when CK goes low and M_1 turns off.

M_2 Turns on so charge $Q_1/2$ appears between both capacitors C_1 and C_2.

Charge is absorbed by M_2 to create channel but is injected back equally between C_1 and C_2 when CK goes high again. So the remaining charge injection on C_2 is $Q_1/2$.

Therefore $\Delta V_{C2} = \frac{\Delta V_{C1}}{2}$ [4]

b) Figure 5.2 shows an autozeroing switched capacitor integrator capable of removing any offset in the amplifier.Derive the output voltage of the integrator during phase Φ1 and Φ2 showing how the offset is removed showing the charge on each capacitor.

[Derivation from notes]

Student should conclude to the following equations showing at each phase how the offset V_{os} gets subtracted.

$$v_{out}^{\Phi 2}(n) = v_{out}^{\Phi 2}(n-1) - (C_1 / C_2) v_{in}^{\Phi 2}(n)$$

$$v_{out}^{\Phi 1}(n) = v_{out}^{\Phi 2}(n) + v_{os}$$

[5]

c) Figure 5.2 shows a schematic of switched current integrator.

i) Derive the transfer function of the switched current integrator and show that in the z-domain this may be represented as H(z)= A/(z-1).[5]

[derivation from notes]

Solutions

6. Figure 6.1 shows a measurement system that digitises the product of two different signals. The architecture uses an operational amplifier, a current multiplying circuit and an analogue-to-digital converter to encode the detected voltage signal. The detected signals are first converted to currents through a voltage-to-current converter (V/I), then multiplied in current mode and finally converted to a spike train through an integrate-and-fire neuron (I/F). The 8 bit counter is used to count the number of spikes in a fixed period T_{reset}. The whole system works off a 1 V supply.

a) Describe a technique which allows detection of very small voltage signals which exist below the 1/f noise floor of the operational amplifier. Include diagrams showing the operation of this technique and how the frequency spectrum of the signal and noise is shaped in each stage.

[5]

We can use the chopper stabilization technique: We up-modulate our input signal using square wave modulation to a frequency which is above 1/f noise spectrum of our amplifier. We then amplify and then down modulate at the output with the same square wave signal. At the output however noise and offset get up-modulated which can then be removed using a low-pass filter.

[2]

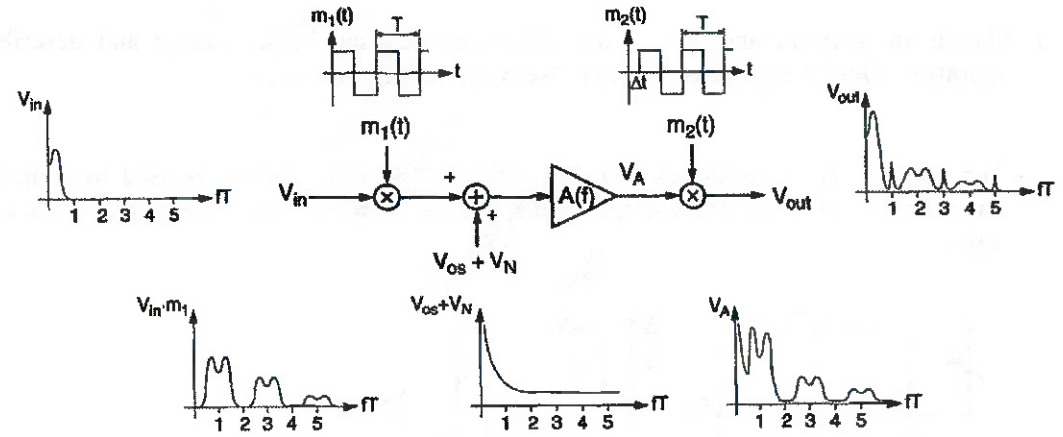


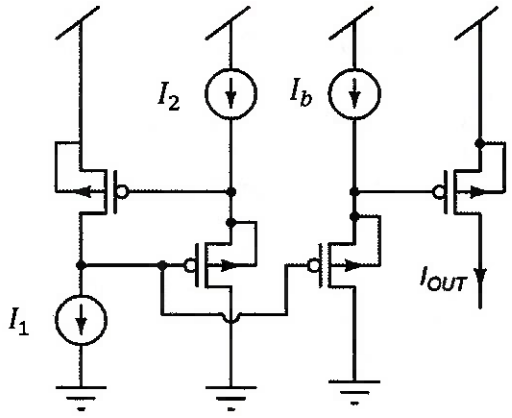
Fig. 9. The chopper amplification principle [19].

[3]

b) Sketch a circuit for multiplying two currents. State the region of operation for the MOSFETs and derive the relationship between the output and the input. Identify a way to provide a gain to multiplication.

[4]

The circuit is a translinear circuit with all MOSFETs operating in weak inversion.



Following the translinear principle,

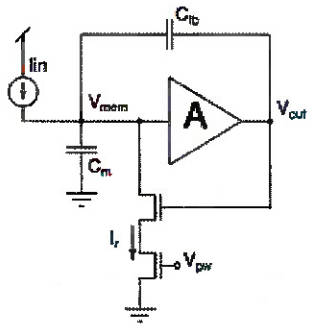
$$I_{out} = \frac{I_1 I_2}{I_b}$$

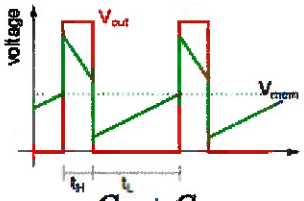
Hence on this circuit, the output current is the multiplication of I_1
The circuit can provide an amplification factor by tuning the value of I_b.

c) Sketch an integrate-and-fire circuit which can be used in this design and describe its operation. Derive equations for the pulse width and frequency.

[5]

We can use an axon-hillock integrate and fire neuron. Current is used to charge the capacitor which resets the voltage after it passes the threshold of the amplifier causing a spike.





$$t_L = \frac{C_{fb} + C_m}{I_{in}} \Delta V_{mem} = \frac{C_{fb}}{I_{in}} V_{dd}$$

$$t_H = \frac{C_{fb} + C_m}{I_r - I_{in}} \Delta V_{mem} = \frac{C_{fb}}{I_r - I_{in}} V_{dd}$$

Frequency $\propto I_{in}$

Pulse width $\propto 1/I_r$ **for** $I_r \gg I_{in}$

[2 points for circuit and explanation, 1point for graph, 2 points for derivations]

[Total 5 points]

d) i. Given that the maximum output is limited to I_{out} = 100 nA, chose a suitable capacitor value for your integrate-and-fire circuit such that the spike frequency is always less than 5 kHz. You may assume the refractory current is I_r = 1 nA.

V_{dd}=1V, I_{in}=100nA, I_r=1nA
$T_{total} = t_n + t_l$ = 1/5KHz
$C_{fd}. V_{dd}/I_{in} + C_{fd}. V_{dd}/(I_r - I_{in})$ = 1/5KHz

C_{fd}= 10pf

[4]

ii. Calculate the maximum allowable reset time T_{rst} such that the counter never overflows for this current. How many bits are required when integrating over a long time constant $T_{rst} = 1\ s$?

[2]

8 bit counter means 2^8 counts therefore Maximum $T_{reset} = 2^8/5x10^3 = 51.2$ msec. [1]
For T_{reset}=1 second we need n=12.28= 13bit counter. [1]

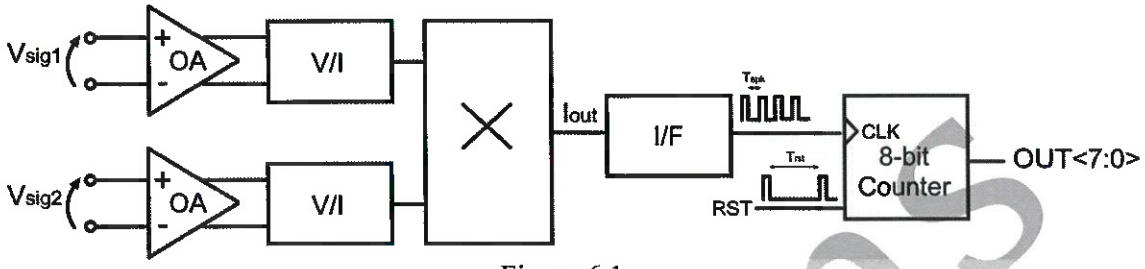


Figure 6.1

Solutions